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Squid Proxy: HTTPS and SSL Bump

Squid is a caching proxy which supports HTTP, HTTPS and FTP. It saves on bandwidth and speeds up response times by caching and reusing frequently requested Web pages. It is multi-platform with built-in access control and, therefore, makes a great server accelerator.

Last month, I had mentioned the use of the Squid proxy server as a way of saving on network data traffic and enjoying a more responsive experience on the Internet. It will be useful as you may be using multiple devices like a laptop, tablet and phones for accessing common sites.

However, examining the access logs of Squid may reveal that there are very few hits. Instead, you are likely to see that a tunnel has been created between the client and the server. This becomes obvious when you realise that most of the sites these days are relying on SSL connections, i.e., HTTPS connections. Hence, any site in the middle is not supposed to know what the content is. Anything else would imply that the security has been compromised!

If SSL connections dominate, with the default Squid environment, an organisation can get all the benefits of a proxy except that there is only a slim chance of reusing objects between different clients.

It is possible to have a trusted group of environments where the loss of privacy by proxy caching data for common use will not be an issue. For such scenarios, Squid provides an *SSL bump* option (<http://wiki.squid-cache.org/Features/SslPeekAndSplice>).

If enabled, the Squid configuration allows the following strategy depending upon the destination server and the client:

- The proxy peeks at the request from the client.
- The proxy establishes a connection with the server on behalf of the client.
- The proxy receives the data and re-encrypts the data using its own certificate.
- The client receives the encrypted data; however, the certificate is not that of the original site but the proxy server's. The user must explicitly trust it.

Configuring Squid for the SSL bump

You will need to create an SSL certificate for Squid

(<http://wiki.squid-cache.org/Features/DynamicSslCert>):

```
cd /etc/squid
mkdir ssl_cert
chown squid:squid
chmod 700 ssl_cert
cd ssl_cert
# Create the certificate
openssl req -new -newkey rsa:2048 -sha256 -days 365 -nodes \
    -x509 -keyout myCA.pem -out myCA.pem
# generate one for importing in a browser
openssl x509 -in myCA.pem -outform DER -out myCA.der
```

The next step is to make changes to the Squid configuration file. The minimal configuration changes required are as follows:

```
# Squid normally listens to port 3128
http_port 3128 ssl-bump generate-host-certificates=on \
    dynamic_cert_mem_cache_size=4MB \
    cert=/etc/squid/ssl_cert/myCA.pem

# bump https traffic
ssl_bump bump all
```

Now, start Squid.

The browsing experience

Enable the use of the proxy in the browsers: Usually, Firefox gives a warning if it comes across a suspicious certificate. I opened Gmail and it warned me; however, it would not allow me to accept a certificate. I needed to explicitly import the certificate *myCA.der* in Firefox, as follows:

Edit 'Preferences'

Click on 'Advanced' section and choose 'Certificates' tab